

Hybrid Reinforcement Learning for Autonomous Driving Using the CARLA Simulator

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Problem Statement

- Collecting real-world driving data is expensive, time-consuming, and safety-critical.
- Testing autonomous driving systems on public roads poses significant safety and regulatory challenges.
- Traditional rule-based driving systems rely on handcrafted logic and struggle to adapt to complex or unseen driving scenarios.
- A realistic simulation environment is required to safely develop, train, and evaluate intelligent autonomous driving agents before real-world deployment.



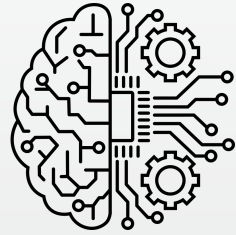
Real-World Motivation

The story that inspired our project

- Companies such as Tesla, Waymo, Mercedes-Benz, and NVIDIA are investing billions of dollars in autonomous driving technology.
- The objective is to reduce accidents, improve traffic efficiency, and enhance passenger safety.
- Developing and testing autonomous driving algorithms directly on public roads is extremely costly and risky.
- Simulation environments such as CARLA enable safe and scalable development before real-world deployment.



Objectives and Applications



Hybrid Learning Framework

Develop a BC + PPO agent



Autonomous Vehicles

Intelligent urban navigation



Autonomous Navigation

Lane Following



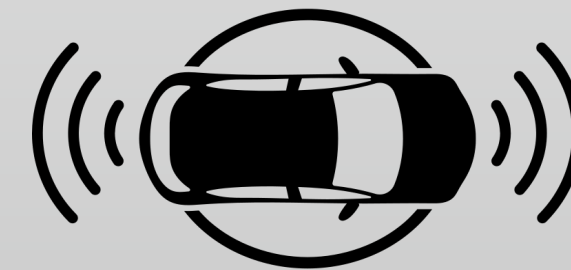
Smart Logistics

Delivery Vehicles



Performance Evaluation

Collision Rate



ADAS

Driver Assistance



Comparative Analysis

Compare BC + PPO

Literature Review

Overview of prior studies on CARLA-based autonomous driving and reinforcement learning

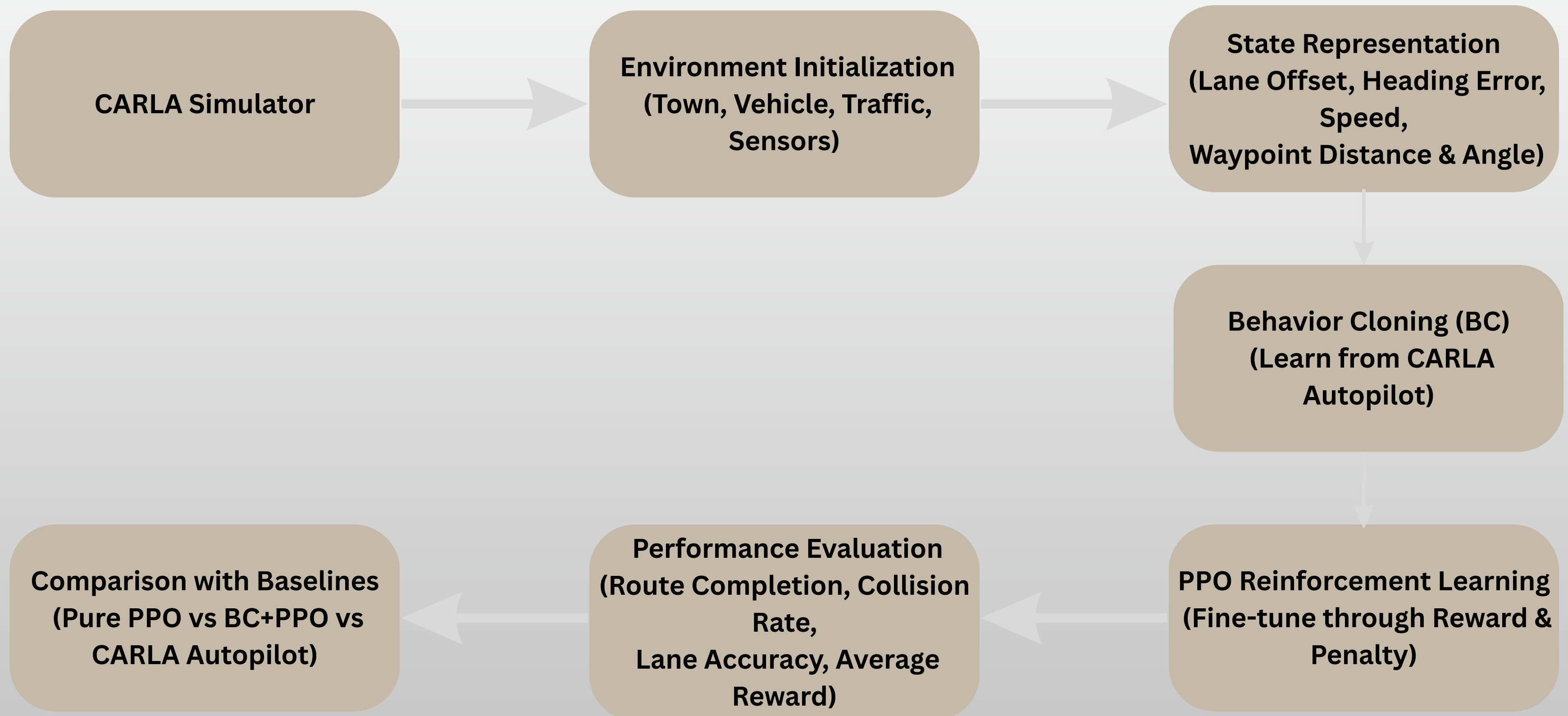
Paper	Approach	Strength	Limitation
Dosovitskiy et al., 2017 – CARLA: An Open Urban Driving Simulator	Modular pipeline vs. IL vs. RL (A3C) benchmarked in CARLA	Established open-source simulator + standard benchmark tasks; enables controlled comparison of approaches	RL (A3C) baseline performs poorly; only 10M training steps used
Schulman et al., 2017 – Proximal Policy Optimization Algorithms	Clipped surrogate objective for policy gradient RL	Stable, simple-to-implement RL algorithm; strong sample efficiency vs. TRPO/A2C/ACER	Not driving-specific; hyperparameters need task-specific tuning
Liang et al., 2018 – CIRL: Controllable Imitative Reinforcement Learning	Pretrain actor via imitation learning, then fine-tune with DDPG + command-conditioned reward branches	Directly validates hybrid IL+RL warm-start approach; trains in ~12 hrs vs. 12 days for pure RL; big gains over CARLA's original RL baseline	Uses DDPG not PPO; still fragile to unseen towns/weather
Chen et al., 2015 – DeepDriving: Learning Affordance for Direct Perception	ConvNet maps images to compact affordance indicators (heading angle, lane distances, car distances) instead of raw pixels or full scene parsing	Introduces the affordance-based structured state representation your project explicitly reuses (lane offset, heading error, waypoint distance/angle)	Predates CARLA (uses TORCS); no RL/imitation hybrid
Chen et al., 2019 – Learning by Cheating	Two-stage: privileged agent (ground-truth map) trained first, then teaches vision-only sensorimotor agent	State-of-the-art results on CARLA benchmark (100% success); shows staged/hybrid training beats direct end-to-end learning	Requires privileged bird's-eye-view info during training; more complex pipeline than BC+PPO

Research Gap Identified:

- No prior work combines Behavior Cloning with PPO specifically for CARLA, CIRL (Liang et al., 2018) pairs imitation learning with DDPG, not PPO.
- Few methods integrate a lightweight affordance based state representation (DeepDriving) with a hybrid BC+RL training pipeline, most either use raw images/BEV maps (Learning by Cheating) or affordance features alone (DeepDriving), not both together.
- Existing CARLA studies benchmark against modular pipelines or pure IL/RL, but there's no direct comparison of Autopilot vs. Pure PPO vs. BC+PPO under identical conditions.

Methodology

Our workflow follows a structured deep learning pipeline – from data collection to explainable AI-based evaluation

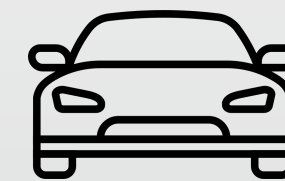


Novelty of the Idea – What Makes Our Project Unique

- Hybrid Learning: Combines Behavior Cloning (BC) with PPO Reinforcement Learning for efficient autonomous driving.
- Structured State Representation: Uses meaningful driving features instead of raw camera images for faster learning.
- Performance Comparison: Compares the proposed model with Pure PPO and the CARLA Autopilot.
- Research Contribution: Investigates whether hybrid learning improves convergence and driving performance.



Hybrid Learning
(Behavior Cloning + PPO)



Structured State
Representation (Affordance-based)



Performance Comparison
Autopilot vs Pure PPO vs Hybrid BC+PPO

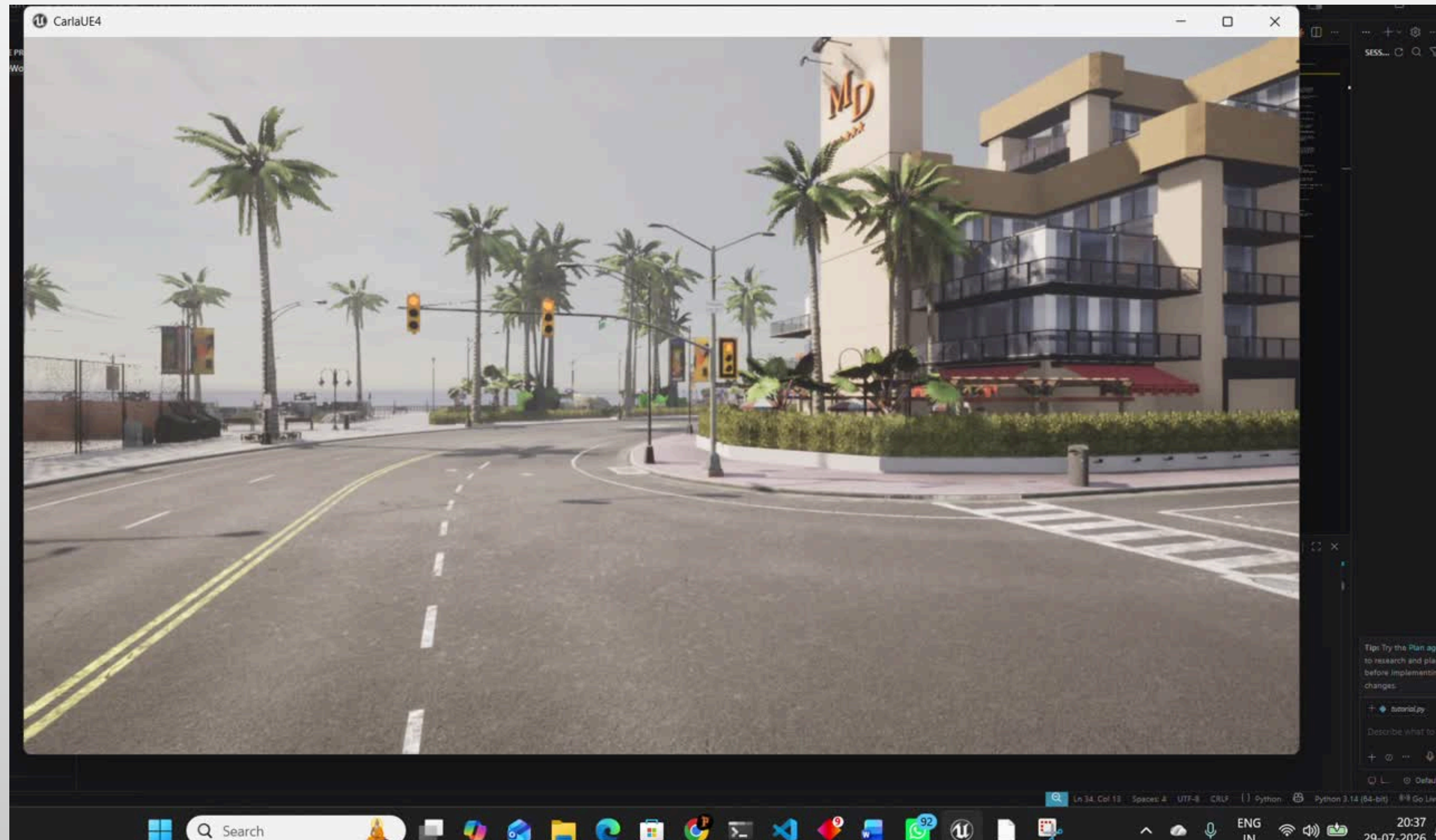


Research Contribution
Efficient & Stable Autonomous Driving

Research Hypothesis: Hybrid BC + PPO will achieve faster convergence and safer navigation than PPO trained from scratch.

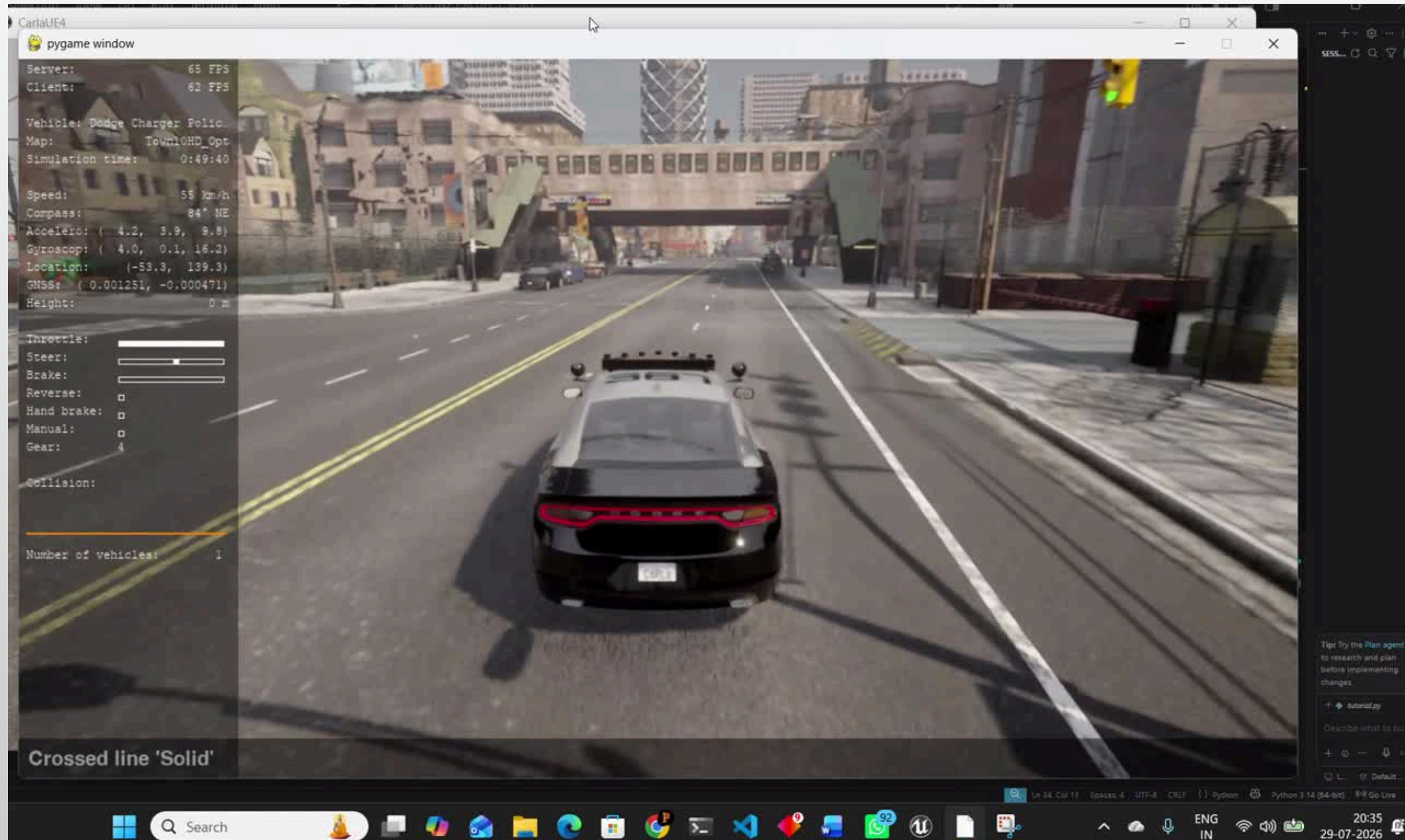
CARLA Simulation Environment

Initial Setup and Environment Validation



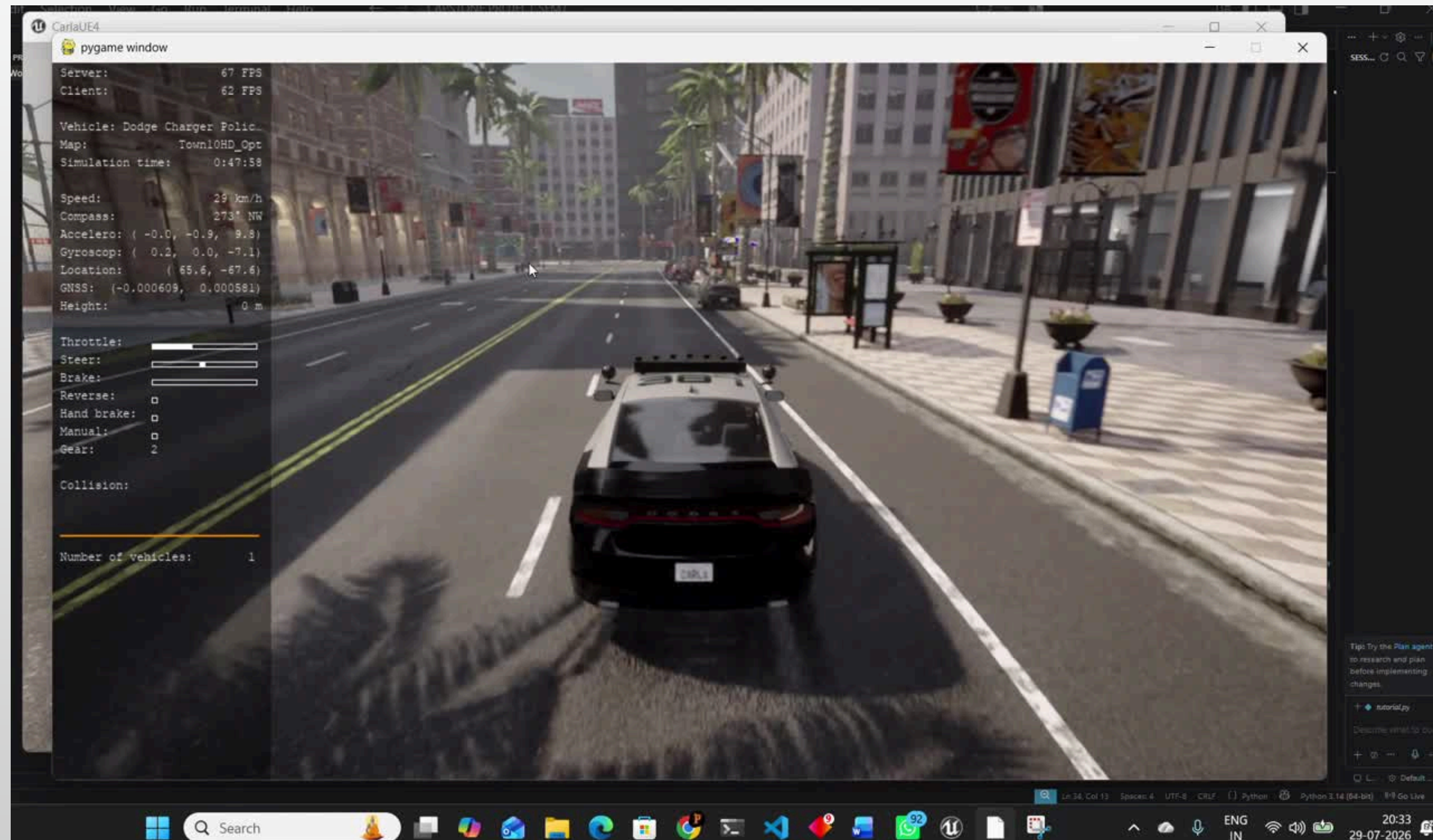
Manual Driving Validation

Verifying Vehicle Control in CARLA



CARLA Built-in Autopilot

Baseline for Comparison



Observation

- Rule-based driving
- No learning capability
- Used as a baseline for evaluation

References

- Dosovitskiy, A., Ros, G., Codevilla, F., Lopez, A., & Koltun, V. (2017). CARLA: An open urban driving simulator. 1st Conference on Robot Learning (CoRL 2017).
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347.
- Liang, X., Wang, T., Yang, L., & Xing, E. (2018). CIRL: Controllable imitative reinforcement learning for vision-based self-driving. European Conference on Computer Vision (ECCV 2018).
- Chen, C., Seff, A., Kornhauser, A., & Xiao, J. (2015). DeepDriving: Learning affordance for direct perception in autonomous driving. International Conference on Computer Vision (ICCV 2015).
- Chen, D., Zhou, B., Koltun, V., & Krähenbühl, P. (2019). Learning by cheating. 3rd Conference on Robot Learning (CoRL 2019).

Tools:

- CARLA Simulator: <https://carla.org/>
- CARLA Documentation: <https://carla.readthedocs.io/>
- CARLA GitHub repository: <https://github.com/carla-simulator/carla>

